

Career Objective

My interest in mobility and intelligent systems was shaped early by the engineering of leading automotive manufacturers like BMW, Honda & Jaguar. I’m keen to work on Automotive Embedded Systems, IoT, and the integration of Artificial Intelligence and Machine Learning in embedded applications.

Skills

Programming Languages: Embedded C, Python, Fullstack Web & App development, AI/ML

System Scripting Tools: Linux, KiCAD Circuit Design, Kubernetes

Analysis: MATLAB, Pandas and Numpy, AWS

Experience

Embedded AI Intern, Spark Minda Limited – Bengaluru, India (*Remote work*) January - July 2026

- Developed a multi-algorithm camera application in C++/Qt that detects lanes, recognizes vehicles, and classifies obstacles.
- A threaded ONNX-YOLO pipeline with a real-time image-processing stage, sustaining 30 FPS at 720p on CPU alone at ~10 W, for the use case of ADAS.

Software Testing Intern, Spark Minda Limited – Bengaluru, India (*On-site*) June – July 2025, February 2026

- Supported development of **UDS (ISO 14229)** services for ECUs in Piaggio Vespa (EU market) using **CAN 2.0 protocol**.
- Utilized **CAPL scripting in CANoe** to simulate & validate diagnostic messages & communication behavior.
- Conducted system testing, analyzed communication logs.
- Built a fully offline Facial Biometric Unlock pipeline in Python for Automotives.
- OpenCV SSD (ResNet-10) , FaceNet 512-d L2-normalised embeddings cutting false accepts to a <0.1 % target at ~5 FPS CPU-only, <2 s unlock, zero network dependency

AI Intern, Tecknodreams Software Consulting pvt. Ltd – Bengaluru, India (*Remote*) June – July 2024

- Built a LinkedIn Resume Optimizer using **RAG** for personalized profile suggestions.
- Integrated embedding and retrieval to align resumes with LinkedIn profiles.
- Delivered streamlined LinkedIn profile enhancements through AI-driven insights & user-document context.

Projects, Certifications and Workshops

CAN Bus–Based Vehicle Telemetry & ECU Simulation November 2025

- Implemented a **two-node CAN bus system using Arduino Uno + MCP2515 modules** to simulate a full vehicle ECU, implementing throttle, braking, indicators, headlight control, fuel modeling, and real-time physics.
- Built a **Python (Pygame) desktop dashboard** that renders a live virtual instrument cluster with RPM, speedometer, odometer, fuel gauge, blinkers, brake indicator, and headlight state **synced over CAN in real time**.
- Tested a structured CAN messaging protocol, fault-tested the bus under sustained load, and validated reliable bidirectional communication **through custom stress tests, echo verification, and dynamic state simulation**.

Campus Sentry – Real-Time Violation & Flood Monitoring Platform October 2025

- Developed a full-stack municipal monitoring system with a **React + Tailwind web dashboard** and a **cross-platform Expo mobile app**, backed by **Firestore, and real-time listeners**.
- Integrated an external **ML pipeline** feeding parking-violation and water-logging detections directly into Firestore, enabling instant approval workflows, dynamic UI updates, and admin dashboards.
- Deployed the web app via **Firestore Hosting** and delivered production-ready Android builds through **EAS, complete with OTA updates, custom app icon, and automated UI splash screen**.

Real-Time Indoor Localization System On Wearable Device using BLE & Machine Learning March 2025

- Built a BLE-based indoor localization system inspired by **IEEE research**, using wearables and **ML**.
- Boosted accuracy with **KNN, SVM, LightGBM, XGBoost, Random Forest**, and more for assistive applications.

IoT-Based Water Quality Monitoring System November 2024

- Designed a prototype using **ESP32 with pH, turbidity, and salinity sensors** and sent the processed data and sent it to the AWS using AWS API, IoT Core, DynamoDB & Lambda Functions.
- Hosted a **Web application** on Firebase to visualize real-time water quality metrics.

Certifications and Workshops

Deep Learning Onramp — MathWorks April 2025

- Completed a training on fundamentals of deep learning, neural networks & model-building in MATLAB.

MATLAB Onramp — MathWorks February 2025

- Completed a certification on MATLAB essentials, data handling, and computational workflows.

- Machine Learning Onramp — MathWorks

February 2025

Completed a course on ML concepts including classification, regression, training pipelines & evaluation techniques.
- Python with AI certification by Coincent.AI

February 2024

Completed a certification on implementing Python in AI through Numpy, Pandas and Matplotlib.
- AI with ML Workshop, IISC Bengaluru

October 2024

Attended a 15-hour workshop by EduFabrica and trained a model to recognize animals using Gradient Descent.
- AR/VR Workshop, Cholalabs

September 2024

Attended a 12-hour workshop at SASTRA University, created a Brick-Breaker game on Blender.

Education

ISEP, Paris	M.Sc IEMDP-Embedded Systems (Ongoing)	September 2026
SASTRA University, Tanjore	B.Tech CSE (Specialization in IoT & Automation) (CGPA : 7.71)	July 2026
DAV Boys Higher Secondary School, Gopalapuram	Secured 95% in XII CBSE Boards.	2022
Indian School Muscat, Oman	Secured 94.4% in CBSE X Boards.	2020

Languages Known

-  Tamil
-  English
-  Hindi
-  French
-  German

Hobbies

-  Bibliophile
-  Digital Art & Comics
-  Flute
-  Carnatic Vocal Singing
-  Basketball